Palaeomagnetic Investigations on the Deccan Traps from Chincholi, Mysore State, India

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Summary – 38 oriented samples of Deccan Traps have been collected from the neighbourhood of Chincholi, Mysore State, India. The Natural Remanent Magnetisation of these rocks has been studied using an astatic magnetometer. It has been found that these rocks are magnetically reversed, the mean magnetic direction being N154°E in declination and 61° down in inclination. Thermoremanance studies conducted on four specimens showed that two specimens with weak NRM and a high secondary magnetisation have Curie temperatures around 560°C for the NRM and exhibited partial reversal of TRM at room temperature, while two specimens with high NRM and with little secondary magnetisation have Curie temperatures much lower than 560° C for the NRM.

1. Introduction

The Palaeomagnetic directions of Deccan Traps of India as were originally measured by Prof. BLACKETT's group from the Imperial College, London, and later by various Indian Scientists, had led to the conclusion that the Indian Landmass had drifted northwards by about 58° of latitude and rotated anti-clockwise by about 36° since the laying down of these traps, about 80 m. years ago. Further the traps showed both 'normal' and 'reverse' magnetic directions. These Traps, situated 2000 ± 200 ft above mean sea level, were found to be normally magnetised, while those occurring below this elevation were reversely magnetised. Thus according to DEUTSCH *et al.* [5][2]), The Traps of India could be reclassified from the sense of their remanent magnetic vector into two classes, namely those occurring above and below the elevation of 2000 ± 200 ft, in contrast to the three-fold geological classification (KRISHNAN [7]) of upper, middle and lower traps. A plausible conclusion of this result seems to be that the earth's field which was opposite to the present direction during the lower Eocene-Upper Cretaceous period flipped over and became 'normal' at a time when the traps above the present elevation of 2000 ± 200 ft erupted.

However, a few cases have been found where the Deccan Traps occurring around the mean sea level elevation showed 'normal' magnetisation. SAHASRABUDHE [10] found 'normal' magnetisation for some Bombay Island lavas (situated at elevation around mean sea level) and sought to explain their normal nature by local geo-

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[2]) Numbers in brackets refer to References, page 123.

logical movements. The Deccan Traps of the East Coast near Rajahmundry in Andhra Pradesh (latitude 17°.03′N and longitude 81°.47′E and elevation around mean sea level) are considered as outlying parts of the main Deccan Traps of Central and Western India, from which they are separated by a distance of about 750 km. They are regarded geologically as belonging to the Lower Traps and showed normal magnetisation (BHIMASANKARAM [1]), while their supposed contemporaries in the main Deccan Traps show 'reverse' directions. If the geological evidence as to the correlation of the Rajahmundry Traps with the lower series of Deccan Traps were correct, either (i) the reverse lower traps of the main Deccan Trap or the Rajahmundry Traps must have acquired their present directions of magnetisation due to some self-reversing mechanism, for which so far there is no supporting evidence or (ii) the earth's field must have flipped over and become normal once during the eruption of the lower traps (i.e.) at a time when the Rajahmundry Traps were erupted, in which case the conclusion that the earth's field was always reversed during the eruption of the Traps occurring below the present elevation of 2000 ± 200 ft should be revised. It appears therefore, that much more detailed investigation of the Deccan Traps, both by extensive sampling at various places, especially where one comes across large number of flows, as well as laboratory studies of the magnetic properties and the petrological characteristics, will be of great interest.

The present work describes palaeomagnetic directions obtained on certain Deccan Traps occurring towards the South-Eastern border of the main Deccan Trap mass and certain studies on their thermoremanance. The samples come from near Chincholi, (Latitude 17°30′N and longitude 77°30′E) in the Gulbarga District of Mysore State.

2. Geology

The area from which the present samples were collected more or less marks the South-Eastern boundary of the Deccan Traps. Being at the fringe of the Deccan Traps, they are believed to represent one of the earliest volcanic activities of the lower Eocene-upper Cretaceous period.

Though the Deccan Traps could be traced South and East of this area, these were found to be highly weathered. The remanent magnetic directions of the rocks from one such area (Shankarapally, Hyderabad District, Andhra Pradesh) were found to be unstable and gave rise to a heavy scatter in their directions.

The Traps in the Chincholi area are overlain by laterite and conglomerate and they overlie the infratrappean unconsolidated beds consisting of white calcareous material, red earth and some clayey material. The geological succession found in this area is a slightly modified version of the general geological succession proposed by MAHADEVAN [9].

Soil	Recent
laterite and conglomerate	Pleistocene and recent
Deccan Traps	Tertiary – Lower eocene

Infratrappeans	Upper Mesozoic to tertiary
Middle Bhimas	Puranas (Vindhyans)
Peninsular granites	Archeans

The Deccan Traps are mainly basaltic in nature and in this area are reported to contain four distinct types viz., columnar, hard, vesicular and amygdoloidal. The vesicular and amygdoloidal Traps were found to be highly weathered and on hammering invariably broke up into small pieces. During the present work, only one variety of the Trap, namely the hard Trap could be successfully sampled.

3. Collection of samples and measurement

During December 1963 about 38 oriented samples of the hard Trap were collected from six sites at a distance of about two miles from one another. From each sample two to three cylindrical specimens of diameter 1″ were cored in the laboratory. The direction and the intensity of remanent magnetisation of the specimens were measured with an astatic magnetometer. Four of the specimens were tested for magnetic stability by thermal-demagnetisation. Six were demagnetised by alternating magnetic fields upto 600 Oe. The thermoremanent characteristics of four specimens were measured in a field of 0.5 Oersted. The variation of saturation remanance (obtained in 3000 Oersteds) with temperature was measured for one sample.

4. Results and discussion

In Fig. 1 are plotted the remanent magnetic directions of all the samples on a Schmidt equal area projection. It shows that for all the samples, the inclination points downwards while for most of the samples the declinations are in the second quadrant with an approximate mean declination of 160°E and mean inclination of 60° down, while the rest of the samples appear to be scattered in the other three quadrants with downward dips not lower than 35°.

A. *Thermal demagnetisation*: Four specimens (two from the cluster in the second quadrant and two from the other quadrants) have been thermally demagnetised so as to test the stability of the magnetisation and to remove any secondary magnetisation.

The instrument used for thermal demagnetisation was developed by one of us (V. L. S. BHIMASANKARAM) in the Physics Department of the Imperial College, London, and will be described elsewhere in detail. It consists of non-inductive non-magnetic vertical cylindrical furnace situated by the side of an astatic Magnetometer. Specimens are inserted into a quartz specimen holder in which a thermocouple is placed, and can be moved up and down and turned through 360° about a vertical axis. The horizontal component of remanance of an oriented cylinder can be measured by turning the specimen through 270° in the usual way, while the vertical component can be measured by moving the holder up and down (COLLINSON and CREER [3]). This

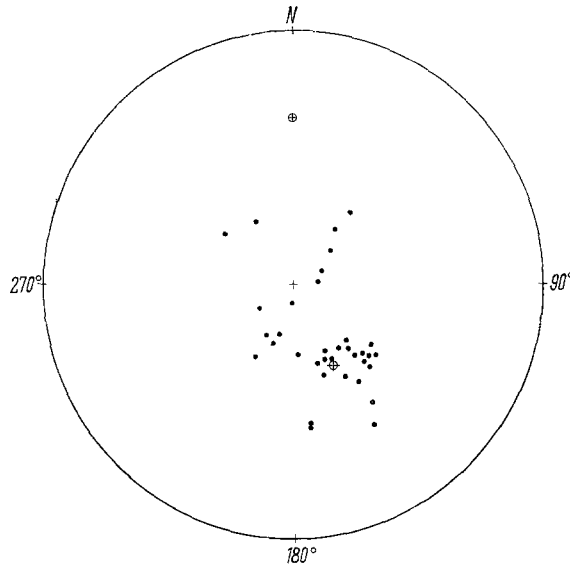


Figure 1
Paleomagnetic directions of samples – Schmidt's equal area projection (all dips down)
⊕ statistical Mean Direction, ⊕ present Dipole

type of instrument eliminates the construction of a specialised furnace and the difficulty of turning the specimen through a horizontal axis by means of a slot in the specimen and a screw driver, to measure the third component of the magnetic moment, as has been done by WILSON [11]. Further, this type of procedure shortens the measuring time considerably. A rock could be heated to 600°C and the variation of magnetic moment observed in 3 perpendicular directions in steps of 25°C in about two hours.

In Figures 2 and 3, are shown the variation with temperature in Zero field of the natural moment of two typical specimens (C 29 and C 37 with declinations in the 3rd and 1st quadrant respectively). In figures 4 and 5 are plotted the variation of directions of the magnetic vectors with temperature in the horizontal and vertical planes for the above two specimens. As can be seen from figures 4 and 5 the room temperature direction of the natural magnetisation of specimen C 29 rotates and oscillates from declination 275°E and inclination 36° upto a temperature of 260°C. Above this it steadies itself and attains a value of 158°E in declination and 58° down in inclination and drops to Zero intensity at the Curie temperature without any further change in directions. Similarly the room temperature direction of the natural remanent vector 68°E and 60° down for specimen C 37 (Fig. 5) rotates upto 140°C and attains a value of 133°E in declination and 65° down in inclination. Above this temperature the vector maintains this direction but decreases to zero at the Curie temperature of 560°C. For two specimens with magnetic directions of 158°E, 63° down (C 20), and 146°E, 52° down (C 30), the original direction of magnetisation did not vary appreciably on demagnetisation.

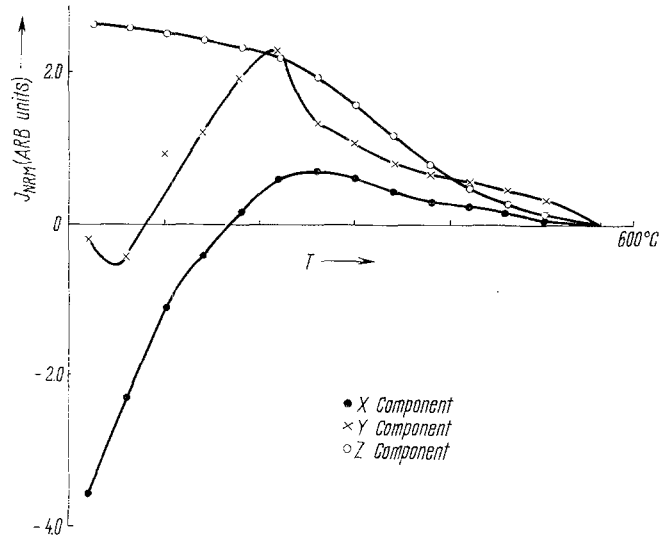


Figure 2
Thermal decay of NRM of specimen C 29 in three perpendicular directions

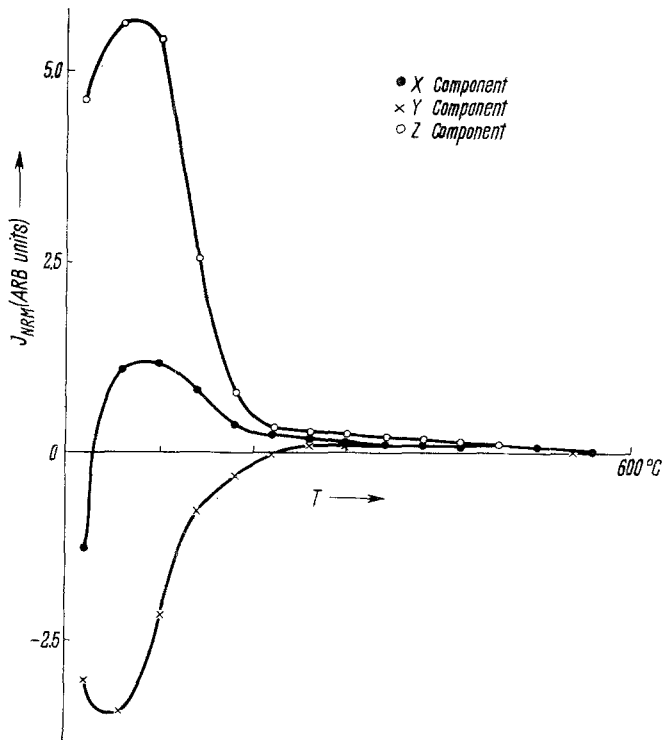


Figure 3
Thermal decay of NRM of specimen C 37 in three perpendicular directions

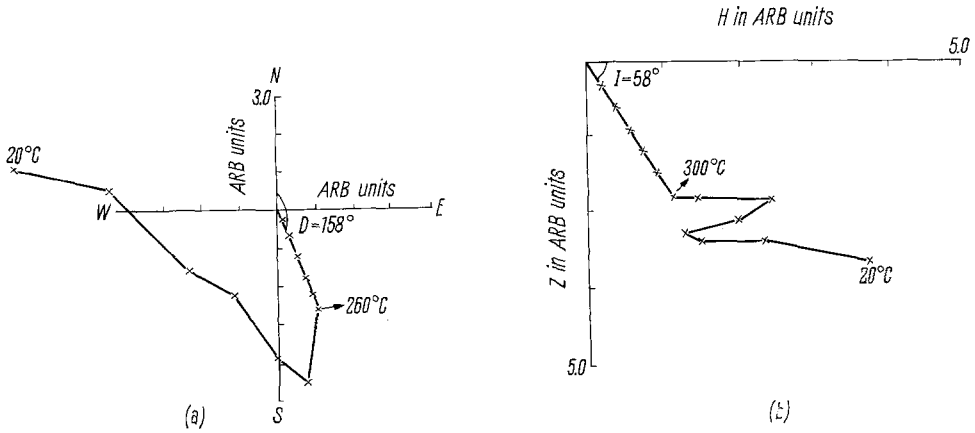


Figure 4
Temperature variation of NRM vector of sample C 29 in (a) horizontal and (b) vertical planes

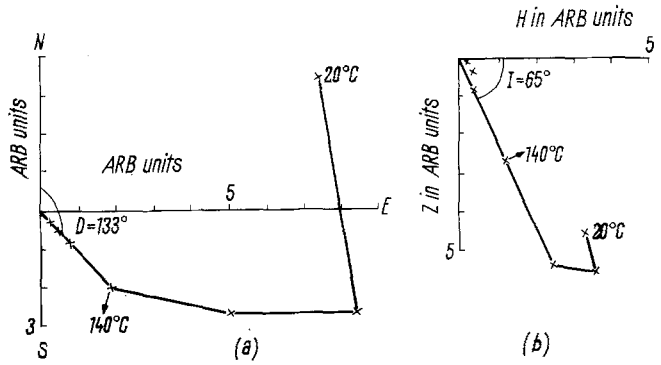


Figure 5
Temperature variation of NRM vector of specimen C 37 in (a) horizontal and (b) vertical planes

The rotation of the magnetic vectors below 260°C for specimen C 29 and below 140°C for specimen C 37, can probably be attributed to secondary magnetisation components while the consistent direction above these temperatures can be taken to be the original remanent magnetic direction obtained by these Traps at the time of their formation. Thus after removal of secondary magnetisation components, both the specimens (C 29 and C 37) showed directions near to the mean direction of the Traps.

B. *A.C. Demagnetisation*: Two specimens (C 19 and C 21) showing high downward dips and nearly zero declination and four specimens showing directions near to the mean value of the group were demagnetised by alternating fields upto 600 Oersteds. These results were obtained by Mr. E. W. McMurry on the instrument he designed in the Imperial College, London. Fig. 6 shows the change in direction experienced by the two anomalous specimens (C 19 and C 21). On demagnetisation

in about 100 to 200 Oersteds, the specimens move towards the mean direction for the group, showing that the secondary magnetisation has been removed. The four specimens having their original directions near the mean direction for the group did not change their direction appreciably, the highest angular change being not more than 5° while for the two anomalous specimens the angular change was more than 45°. The original intensity of NRM of the specimens C 19 and C 21 is of the order of 7×10^{-4} emu/gm., while for the other four specimens it is 5 to 7 times greater.

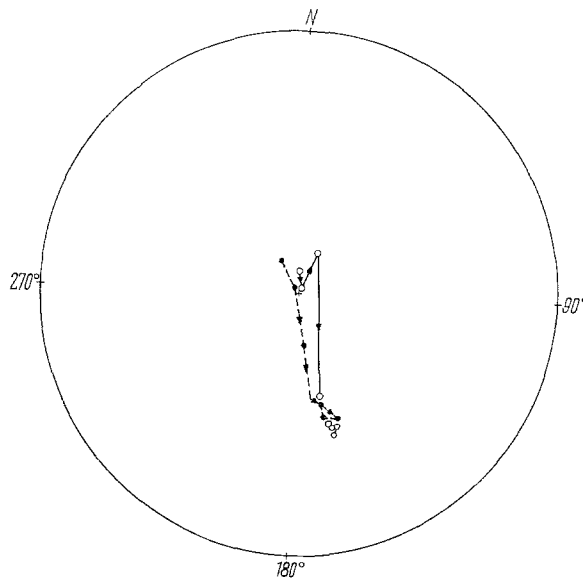


Figure 6
Rotation of the remanent magnetic direction on A.C. demagnetisation (All dips down) ○–C 19; ●–C 21

The variation of the ratio of the intensity at any stage of demagnetisation to that of the original NRM (J_H/J_0) is plotted in Fig. 7 and shows a more or less smooth decrease with increasing demagnetisation field, falling to about 10% at 600 peak Oersteds.

Thus the results of a.c. demagnetisation show that these specimens with original directions near to the mean value are magnetically stable while those with original directions much different from the mean value have certain amount of secondary magnetisation which could be removed by a.c. demagnetisation.

C. *Mean magnetic direction of the rocks*: The mean remanent magnetic direction of the rocks, calculated by the method described by FISHER [6], was found to be 154°E in declination and 61° down in inclination. This mean direction is taken as representative direction of the Traps of this area and differs very much from the present declination and dip of the earth's field. This can be explained by the assumption that the rocks were magnetised when the geomagnetic field was reversed, and that since their formation the Indian sub-continent has moved northward across the geographic

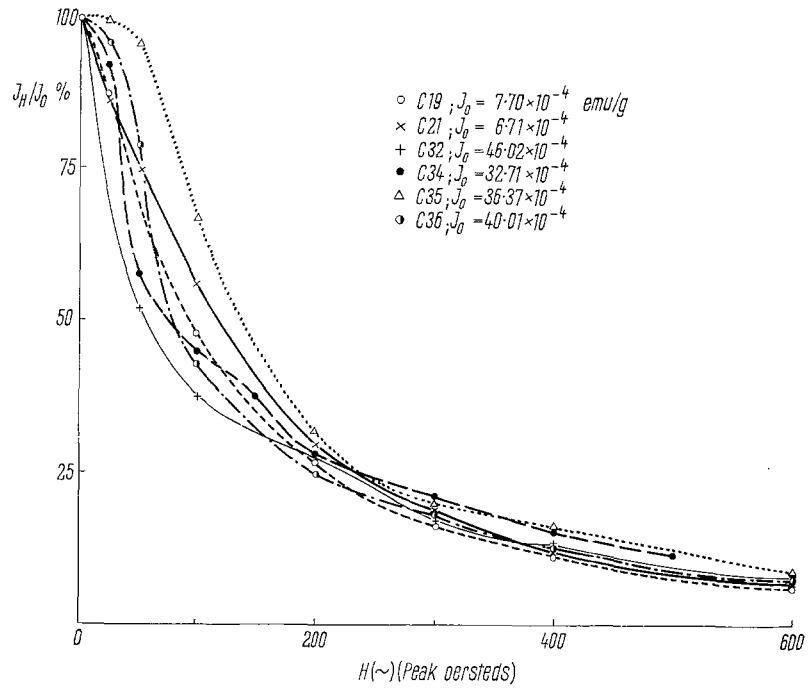


Figure 7
Variation of NRM with alternating demagnetisation field

equator and rotated in an anti-clockwise direction. This result is in agreement with those of earlier workers on the Deccan Traps (CLEGG *et al.* [2] and DEUTSCH *et al.* [4]), see for instance the Amba-lower traps with declination of 144°E and dip of 60° down, from the South-Western border of the Deccan Traps). The ancient latitude (λ) of the site at the time of formation of these Traps could be calculated on the assumption that the magnetic direction of the rocks corresponds to that of a geocentric magnetic dipole lying along the earth's rotational axis, so that $\tan \lambda = \frac{1}{2} \tan I$, where I is the inclination of the rocks. The ancient latitude obtained in this case is 42° south and is in agreement with the ancient latitudes for the Amba and Nipani sites when the lower most traps erupted there. Thus the drift of the present site is about 60° of latitude (North-wards) and the anti-clockwise rotation is about 26°. The magnetic pole at the time of eruption of these traps was calculated to be at 26°N latitude and 81°W longitude.

D. *Thermoremanent magnetisation*: The thermoremanances of 4 samples (C 20, C 30, C 29, C 37) obtained by cooling the samples in 0.5 Oe. from about 600°C to room temperature were obtained with the object of studying the nature of acquisition of natural magnetisation. In Fig. 8a to d are shown the reheating (in zero field) curves of the TRM obtained in 0.5 Oersted, along with the decay of their NRM on heating them for the first time in zero field. The salient results from these observations have

Table 1

S. No.	$J_{NRM \times 10^4}$ emu/gm	$J_{TRM\ 0.5/Oe}$ at 20°C emu/gm	J_{TRM}/J_{NRM}	Curie temperatures				Remarks
				NRM 1	NRM 2	TRM 1	TRM 2	
C 20	100.0	44.0	0.44	340°C	–	340°C (?)	540°C	Strong NRM with little or no secondary magnetisation
C 30	145.0	52.0	0.36	180°C	–	180°C (?)	540°C	
C 29	4.5	30.0	6.66	180–220°C (?)	560°C	180–220°C (?)	560°C	Weak NRM with secondary magnetisation shows partial reversal in TRM
C 37	5.6	20.0	3.57	(?)	560°C	140°C	560°C	

been tabulated in Table 1. It will be observed from the above figures and the table that for two specimens C 20 and C 30, the Curie temperatures of NRM are 340°C and 180°C respectively while for the other two they are about 560°C, though lower Curie temperatures abou t180°C–220°C, might be also present for the latter two specimens. The intensity of NRM at room temperature for the first two specimens (C 20 and C 30) is 20 to 30 times that for the specimens C 29 and C 37. It is also interesting to note that these two specimens with high NRM and lower Curie temperatures (340°C for C 20 and 180°C for C 30) have their original NRM directions nearer to the mean direction of this group of rocks. Hence they have very little secondary magnetisation while the other two specimens (C 29 and C 37) with weak NRM and Curie temperature around 560°C show NRM directions much different from the mean direction indicating that they had acquired secondary magnetisation, which was removed by thermal demagnetisation (Figs 4 and 5).

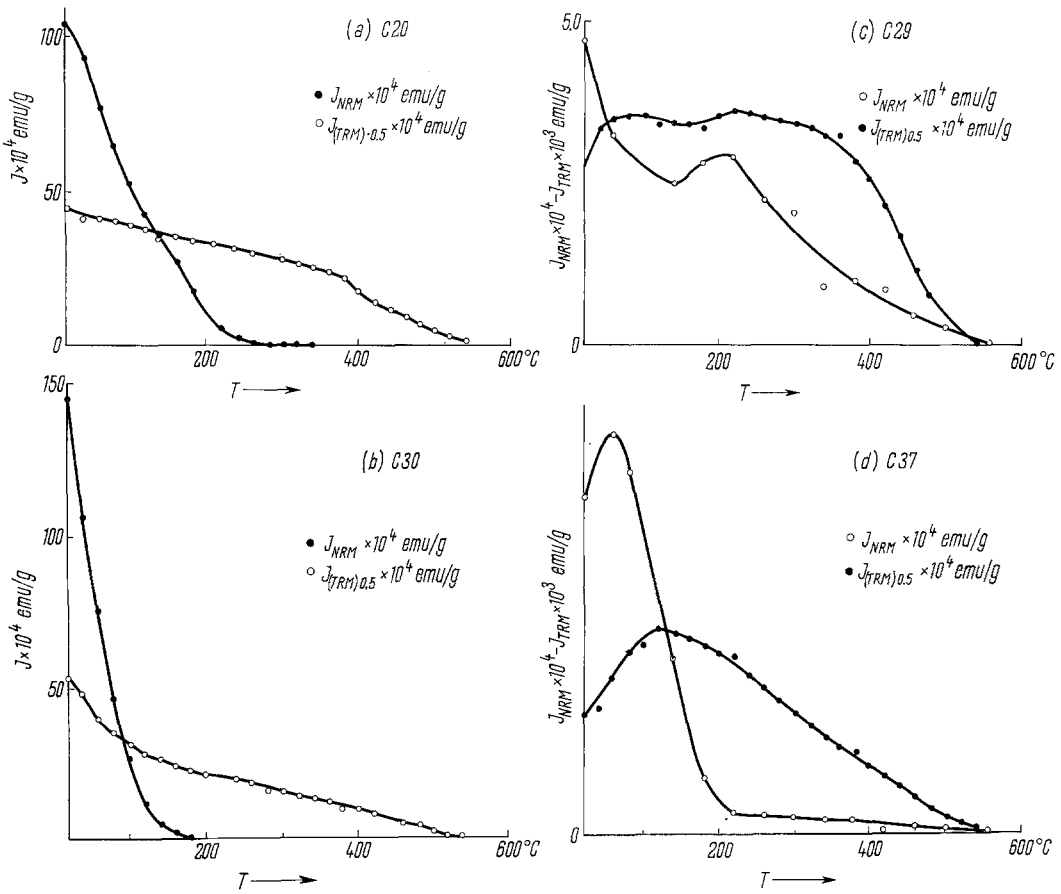


Figure 8
Thermal decay of J_{NRM} and J_{TRM} 0.5 oe in zero field

However, the TRM curves for all specimens showed the higher Curie temperature of about 540/560°C and possibly another lower Curie temperature below 340°C. (See Table 1 and Figs. 8a to d). Specimens C 29 and C 37 with weak NRM and high Curie temperature of 560°C showed partial reversal phenomenon in the TRM curve. The partial reversal of TRM for specimen C 37 (Fig. 8d) is a similar type observed for some Linga Traps (Chindwara District, Madhya Pradesh) by LEWIS [8].

While the thermoremance data described here are too meagre to warrant definite conclusions, they broadly bring out the following facts:

(i) Two specimens with strong NRM and Curie temperature much lower than 560°C have little or no secondary magnetisation.

(ii) Two specimens with weak NRM and Curie temperature around 560° have high secondary magnetisation and show some amount of partial reversal of the TRM at the room temperature.

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